Validation status and source guide

Prototype; not yet carefully vetted. The figures, derived tables and reference code can support discussion and follow-up. They are not final publication results.

The latest presentation revision aligns panels a and c at 110 mm, makes DOF bold, and sets each matrix cell to about 7.1 mm with 8.5 pt numbers and about 1.46 mm side padding. Panel d now occupies the lower-right slot, retaining all 180 pairs with larger nodes and labels. Rendering and arithmetic checks do not establish scientific validity.

Specific source issues and repair gates

Endodermis: three fer-4 contexts have filtered graphs that disagree with their stored outgoing-degree scores. First compare the filtered graphs with the existing raw links and declared filter. Recompute scores from the confirmed graph in isolated working directories if needed. A fresh inference fit is not yet shown to be necessary. If scores change, rerun nominations, enrichment and GRN-dependent overlaps.

Phloem and xylem: substring-based target-feature selection admitted genes selected from pole-pericycle DE contexts into the separate tissue fits. This concerns gene features, not accidental inclusion of pericycle cells. Correcting these models requires an explicit map and controlled refitting; deleting edges afterward is not equivalent.

Interpretation: nominations reflect relative network ranks, not measured TF activity. Context unions do not demonstrate co-occurrence within one cell or intercellular transport. YUC5 was excluded before fitting in several tissues, including endodermis. Its missing edges there are not negative biological evidence.

Audit status

The requested independent audit report is available: AUDIT_REPORT.md and verification and wording correction. Its scope and unresolved findings must be read before reuse. Audit completion is not publication approval.

Detailed evidence: validation status; source availability; endodermal triage.

Available data and code

The pinned paper GitHub repository contains analysis code and exported tables. Integrated scRNA-seq, CellOracle input/raw-link/filtered-link objects and full DE tables are present on Resnick. Exact remote paths, check dates and limits are in the linked source record; large objects were not all reloaded for this layout revision.

The complete input/output/code inventory records exact repository-relative paths, calculation and plotting scripts, outputs, commands and dependencies for all six panel types. Current presentation notes identify the selected panel versions and the new panel d.

The code reference README and source/copy hash manifest accompany 30 reference scripts, each labelled prototype. They are source snapshots with original project dependencies, not a self-contained portable pipeline. Current plotting code is linked on the following pages. No GitLab release has been made.

Additive reproducibility check: membership reconstruction R script and its comparison report. This reproduces memberships from the recorded nomination and DE tables; it does not repair upstream network scores.

Panel inputs, outputs and code

Links open local files in this checkout. Original source paths and hashes are retained in the CSV inventory. Code snapshots are explicitly marked prototype.

a: Top-ten TF-family forest | 110 x 80 mm

Inputs: source_top10_family.csv

Calculation: Fisher enrichment from distinct nominated TFs against paired-score Kay TF background; retain PlantTFDB definition; select ten families with >=10 eligible members by max WT/fer4 odds ratio. Preserve 95% two-sided Fisher CI and one-sided enrichment P, BH across 55 families per genotype. Plot existing numbers only.

Plotting code: make_review_v5_forest.R
Output: top10_aligned_forest_review_v5.pdf (PDF; PNG and editable SVG are saved alongside)

Scope: 33 eligible DOFs differs from 37-member family list. Conditional on prior and scored universe; not genotype-difference test. Pooled score nominations inherit unresolved endodermal score/graph and phloem/xylem feature-selection findings.

b: DOF evidence bars | 85 x 46 mm

Inputs: dot_membership.csv
fig6_full_intersections.csv

Calculation: Count 37 DOFs by nomination and DEG status: 12 both, 14 nominated without DEG, 0 DEG without nomination, 11 neither. Independently assert these four counts against the saved complete intersection table. Stack starting positions are derived from within-group cumulative counts in v6; numerical data remain unchanged.

Plotting code: make_dof_evidence.R
Output: dof_evidence_bars.pdf (PDF; PNG and editable SVG are saved alongside)

Scope: Conditional view of the full 37-member DOF family. DE and network nomination are not independent evidence branches. The original standalone membership generator was not located. A new R reconstruction reproduces the gene identities and flags from the recorded source tables; display-symbol differences are preserved and reported.

c: Ten-DOF cell-type matrix | 110 x 90 mm

Inputs: celltype_summary_all37.csv

Calculation: Build 37 x 44 gene-context statuses, count WT/fer4 nominations by cell identity, divide by zones scored in both genotypes for the same gene. Select ten assayed DOFs. The v6 plotting script reads only celltype_summary_all37.csv; no TF-family plotting table is read.

Plotting code: make_review_v6_matrix.R
Output: celltype_tested10_review_v6.pdf (PDF; PNG and editable SVG are saved alongside)

Scope: Descriptive context counts, not replication, TF activity, or a cell-type-only fitted network. Opposite directions can occur in different zones. Original pooled score nominations and missingness are preserved, including unresolved source findings.

Panel inputs, outputs and code (continued)

Links open local files in this checkout. Original source paths and hashes are retained in the CSV inventory. Code snapshots are explicitly marked prototype.

d: All-context DOF-auxin network | 135 x 90 mm

Inputs: all_contexts_as_exported_union_edges.csv

Calculation: Read all-ten-tissue tested-DOF edges, normalize target symbols where the annotation table supplies them, select IAA/SAUR/GH3/TAA/TAR/YUC names, collapse source-target pairs across contexts/genotypes. Compare pairs with seven-fit reference. 935 source rows ->180 pairs, 10 DOFs, 37 targets; selected edges in 43 contexts/83 genotype-contexts. 147 pairs have seven-fit support, 33 occur only in excluded fits. V7 changes display geometry, node sizes and typography only.

Plotting code: make_panel_d.R
Output: all_contexts_panel_d.pdf (PDF; PNG and editable SVG are saved alongside)

Scope: As-exported descriptive union, not one jointly fitted graph. Includes all known issue classes; support categories do not establish which edges survive corrected fits. Symbol-family filter is not pathway enrichment or complete auxin biology. Positions are geometry only; no activation, repression or intercellular transport inferred.

Supporting: Vascular-context network | 140 x 75 mm

Inputs: tested10_retained_edges_seven_unaffected_consistent_tissues.csv
clean_programme_target_edges.tsv

Calculation: Apply same target-symbol family filter, retain elongating phloem-pole pericycle, xylem-pole pericycle and procambium, and union WT/fer4 source-target pairs. 207 source rows ->81 pairs, 10 DOFs and 23 targets in 3 contexts.

Plotting code: make_overviews.R
Output: vascular_overview_compact.pdf (PDF; PNG and editable SVG are saved alongside)

Scope: Illustrative selected vascular subset. Excludes endodermis/phloem/xylem tissue fits; pole-pericycle contexts are pericycle, not those excluded tissue fits. Union is neither a single fitted graph nor a spatial-transport model. No pooled coefficient or DE encoding.

Alternative b: Earlier compact UpSet | 85 x 55 mm

Inputs: fig6_full_intersections.csv

Calculation: Select exclusive multi-set intersections 111,110,101,011 from seven-region table:12,14,0,172. Assert union 5805 and set totals 37,512,5466. Omit singleton bars (11,314,5282), retain complete source table.

Plotting code: make_compact_upset.R
Output: compact_upset.pdf (PDF; PNG and editable SVG are saved alongside)

Scope: Exclusive, not inclusive intersections. DOF+DEG without GRN=0; inclusive DOF-DEG overlap=12. Three singleton regions are omitted from the plotted rows. A new R reconstruction reproduces all seven exclusive counts from the recorded source inputs. Network-dependent counts inherit score-source caveats.